

Tutorial Sheet 2

Optimal Control via Dynamic Programming

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Exercise 1. Consider a scalar, discrete-time system (*i.e.* the state $x \in \mathbb{R}$) with dynamics

$$x_{i+1} = x_i + u_i,$$

where $u_i \in \mathbb{R}$ denotes the control input and can take one of three distinct values, namely $u \in \{-1, 0, 1\} \triangleq \mathcal{U}$. Consider the time horizon $k = 0, \dots, N$, for some $N \in \mathbb{Z}_{\geq 0}$, and suppose we wish to determine $u(i)$, for $i = 1, \dots, N$ such that the cost

$$J(u, x_0) = \sum_{i=0}^{N-1} (ax_i^2 + bu_i^2) + cx_N^2,$$

with $a \geq 0$, $b \geq 0$ and $c \geq 0$, is minimised and such that $x_N = 0$.

Suppose $a = 1$, $b = 0$, $c = 1$ and $N = 3$. Determine the optimal policy that achieves this using the Bellman equation.

$$V(x, 3) = 0$$

$$V(x, 2) = \min_{u_1 \in \mathcal{U}} \{ L(x_1, u_1) + V(x, 3) \}$$

$$= \min_{u_1 \in \mathcal{U}} \{ x_1^2 \}$$

$$\therefore u_2^* = \begin{cases} 1 & , \text{ if } x_2 = -1 \\ 0 & , \text{ if } x_2 = 0 \\ -1 & , \text{ if } x_2 = 1 \end{cases}$$

$$\text{Therefore, } V(x, 2) = \begin{cases} 1 & , \text{ if } x_2 = -1 \\ 0 & , \text{ if } x_2 = 0 \\ 1 & , \text{ if } x_2 = 1 \end{cases}$$

$$V(x, 1) = \min_{u_1 \in U} \{ x_1^2 + V(x, 2) \}$$

$$\therefore u_1^* = \begin{cases} 1, & x_1 = -2 \\ 1, & x_1 = -1 \\ 0, & x_1 = 0 \\ -1, & x_1 = 1 \\ -1, & x_1 = 2 \end{cases}$$

$$\text{Therefore, } V(x, 1) = \begin{cases} 5, & x_1 = -2 \\ 1, & x_1 = -1 \\ 0, & x_1 = 0 \\ 1, & x_1 = 1 \\ 5, & x_1 = 2 \end{cases}$$

$$V(x, 0) = \min_{u_1 \in U} \{ x_0^2 + V(x, 1) \}$$

$$\therefore u_1^* = \begin{cases} 1, & x_0 = -3 \\ 1, & x_0 = -2 \\ 1, & x_0 = -1 \\ 0, & x_0 = 0 \\ -1, & x_0 = 1 \\ -1, & x_0 = 2 \\ -1, & x_0 = 3 \end{cases}$$

$$\text{Therefore, } V(x, 0) = \begin{cases} 14, & x_0 = -3 \\ 5, & x_0 = -2 \\ 1, & x_0 = -1 \\ 0, & x_0 = 0 \\ 1, & x_0 = 1 \\ 5, & x_0 = 2 \\ 14, & x_0 = 3 \end{cases}$$

Exercise 2. Consider a dynamical system

$$\dot{x} = \tilde{f}(x, u),$$

where x and u denote the state and control input, respectively, and $\tilde{f}$ is a sufficiently smooth mapping. Suppose we wish to design the control input u to minimise the cost functional

$$J(u, x_0) = \int_0^T L(x(\tau), u(\tau)) d\tau + \Phi(x(T)),$$

where L and Φ are sufficiently smooth.

Suppose $u^*(x(t))$, for all $t \in [0, T]$ is an optimal policy. Show that the subpolicy

$$u_s^*(x(t)) = u^*(x(t)),$$

for all $t \in [s, T]$, with $0 \leq s \leq T$, is optimal with respect to the cost

$$J(u_s, x_s) = \Phi(x(T)) + \int_s^T L(x(\tau), u(\tau)) d\tau.$$

We can prove this by contradiction.

Assume for sub-policy $\tilde{u}_s(x(t))$, which

$$J(\bar{u}_s, x_s) = \int_s^T L(x(\tau), \bar{u}_s(\tau)) d\tau + \Phi(x(T)) < J(u_s^*, x_s)$$

Therefore,

$$\begin{aligned} J^*(u, x) &= \int_0^T L(x(\tau), u^*(\tau)) d\tau + \Phi(x(T)) \\ &= \int_0^s L(x(\tau), u^*(\tau)) d\tau + \int_s^T L(x(\tau), u_s^*(\tau)) d\tau + \Phi(x(T)) \\ &= \int_0^s L(x(\tau), u^*(\tau)) d\tau + J(u_s^*, x_s) \\ &\stackrel{>}{\circlearrowleft} \int_0^s L(x(\tau), \bar{u}_s(\tau)) d\tau + J(\bar{u}_s, x_s) \end{aligned}$$

which contradicts the statement that $u^*(x(t))$ is optimal policy.

Therefore, for optimal control policy $u^*(x(t))$, sub-policy $u_s(x(t))$ is optimal.

Exercise 3. Consider a scalar (*i.e.* the state $x \in \mathbb{R}$), continuous-time system that evolves according to the dynamics

$$\dot{x} = kx(t)^3 + u(t),$$

where $u \in \mathbb{R}$ is the control input and $k > 0$ is a constant parameter. Suppose we wish to design the input u to minimise the cost functional

$$J(u, x_0) = \frac{1}{2} \int_0^\infty \alpha x^6 + ru^2 dt,$$

where $\alpha > 0$ and $r > 0$ are constants.

- (a) Show that the HJB equation associated with this infinite-horizon control problem optimal is an ordinary differential equation given by

$$kx^3 \frac{dV}{dx} - \frac{1}{2r} \left(\frac{dV}{dx} \right)^2 + \frac{1}{2} \alpha x^6 = 0.$$

- (b) Show that the value function associated with the problem is given by

$$V(x) = \frac{r}{4} \left(k + \sqrt{k^2 + \frac{\alpha}{r}} \right) x^4.$$

- (c) What is the optimal control for the considered problem?
 (d) Explain why

$$V(x) = \frac{r}{4} \left(k - \sqrt{k^2 + \frac{\alpha}{r}} \right) x^4$$

is *not* a value function.

$$(a) \quad H(x, u, \lambda) = L(x(t), u(t)) + \lambda^T f(x, u)$$

$$\therefore H(x, u, \frac{\partial V}{\partial x}) = \frac{1}{2} (\alpha x^6 + ru^2) + \frac{\partial V}{\partial x} (kx^3 + u)$$

$$\therefore \frac{\partial H}{\partial u} = ru + \frac{\partial V}{\partial x}$$

$$\frac{\partial^2 H}{\partial u^2} = r > 0, \text{ corresponds to minimum}$$

$$\text{for } \frac{\partial H}{\partial u} \Big|_{u=u^*} = 0, \quad u^* = -\frac{1}{r} \frac{\partial V}{\partial x}$$

$$\begin{aligned}\therefore H^*(x, \frac{\partial V}{\partial x}) &= \frac{1}{2} (ax^6 + \frac{1}{r} (\frac{\partial V}{\partial x})^2) + \frac{\partial V}{\partial x} (kx^3 - \frac{1}{r} \frac{\partial V}{\partial x}) \\ &= kx^3 \frac{\partial V}{\partial x} - \frac{1}{2r} (\frac{\partial V}{\partial x})^2 + \frac{1}{2} ax^6\end{aligned}$$

$$\therefore -\frac{\partial V}{\partial t} = H^*(x, \frac{\partial V}{\partial x}) = 0$$

$$\therefore kx^3 \frac{\partial V}{\partial x} - \frac{1}{2r} (\frac{\partial V}{\partial x})^2 + \frac{1}{2} ax^6 = 0$$

(b) for $V(x) = \frac{r}{4} (k + \sqrt{k^2 + \frac{a}{r}}) x^4$

$$\frac{\partial V}{\partial x} = r (k + \sqrt{k^2 + \frac{a}{r}}) x^3$$

$$\therefore kx^3 \frac{\partial V}{\partial x} - \frac{1}{2r} (\frac{\partial V}{\partial x})^2 + \frac{1}{2} ax^6$$

$$= kr (k + \sqrt{k^2 + \frac{a}{r}}) x^6 - \frac{1}{2} r (2k^2 + \frac{a}{r} + 2k\sqrt{k^2 + \frac{a}{r}}) x^6 + \frac{1}{2} ax^6$$

$$= 0$$

(c) $u^* = -\frac{1}{r} \frac{\partial V}{\partial x} = -r (k + \sqrt{k^2 + \frac{a}{r}}) x^3$

(d) Because $V(x)$ might < 0

(b) let $\frac{\partial V}{\partial x} = D$. $\frac{1}{2} ax^6 + kDx^3 - \frac{1}{2r} D^2 = 0$

Therefore, the roots can be easily found by:

$$D = \frac{-kx^3 \pm \sqrt{k^2 x^6 + (a/r) x^6}}{-(-1/r)}$$

$$= r (k \pm \sqrt{k^2 + \frac{a}{r}}) x^3$$

$$\therefore V = \frac{1}{4} r (k \pm \sqrt{k^2 + \frac{a}{r}}) x^4 + C$$

$$\therefore V(0) = 0$$

$$\therefore C = 0$$

$$\therefore V > 0 \text{ for } x > 0$$

$$\therefore V = \frac{r}{4} \left(k + \sqrt{k^2 + \frac{a}{r}} \right)$$

(d) This would make $V < 0$ for $x > 0$, which not correspond to $J > 0$ for $x > 0$

Exercise 4. Suppose you administer a certain resource, that evolves over time according to the dynamics

$$\dot{x} = ax(t) - u(t),$$

where $a > 0$ represents a constant growth rate and $u(t)$ represents your rate of consumption of the resource. You are to select your consumption rate so as to minimise a cost functional of the form

$$\min_{u(t)} J(u(\cdot), x_0) = \int_0^T -e^{-\rho t} \ln(u(t)) dt,$$

where $\rho > 0$ is a constant parameter.

- (a) What is the Hamiltonian associated with this optimal control problem?
- (b) What is the optimal Hamiltonian associated with the optimal control problem?
- (c) What is the HJB equation associated with the optimal control problem? Remember to include the available boundary condition on the value function.
- (d) Determine conditions on the time-varying functions $c(t)$ and $d(t)$ such that

$$V(x, t) = -e^{-\rho t}(c(t) + d(t) \ln(x))$$

is the value function associated with the considered optimal control problem.

- (e) Determine an expression for the optimal control.
- (f) In the following sub-tasks, you will use Matlab (or your preferred alternative “tool”) to determine $c(t)$ and $d(t)$, based on the conditions derived in part (d) above and to determine some other “properties” of the optimal control problem. Towards this end, consider the setting in which $a = 1$ and $\rho = 1$.
 - (i) Let (at least at this initial stage), $T = 10$. Using, for instance, ode45, obtain a numerical representation of the functions $c(t)$ and $d(t)$ for $t \in [0, T]$.
 - (ii) We will not let T vary, from $T = 1$ to $T = 100$. Plot the optimal-cost-to-go as a function of the terminal time T .
 - (iii) Explaining your reasoning, provide an estimate for what the optimal cost-to-go corresponding to the infinite-horizon problem (*i.e.* $T \rightarrow \infty$) is. This can be determined without relying on any simulations. However, you can run simulations to see that the theory represents what you actually observe in simulation!

$$(a) \quad H(x, u, \lambda) = L(u, x) + \lambda^T \tilde{f}(x, u)$$

$$\therefore H(x, u, \frac{\partial V}{\partial x}) = -e^{-\rho t} \ln(u) + \frac{\partial V}{\partial x}(ax - u)$$

$$(b) \quad \frac{\partial H}{\partial u} = -\frac{e^{-\rho t}}{u} - \frac{\partial V}{\partial x}$$

$$\frac{\partial^2 H}{\partial u^2} = \frac{e^{-\rho t}}{u^2} > 0, \text{ correspond to minimum}$$

$$\text{for } \frac{\partial H}{\partial u} \Big|_{u=u^*} = 0, \quad u^* = -\frac{e^{-\rho t}}{\partial V / \partial x}$$

$$\begin{aligned}
\therefore H^*(u, \frac{\partial V}{\partial x}) &= -e^{-\rho t} \ln(-\frac{e^{-\rho t}}{\partial V / \partial x}) + \frac{\partial V}{\partial x} (ax + \frac{e^{-\rho t}}{\partial V / \partial x}) \\
&= -e^{-\rho t} (-\rho t - \ln(-\frac{\partial V}{\partial x})) + ax \frac{\partial V}{\partial x} + e^{-\rho t} \\
&= e^{-\rho t} (\rho t + \ln(-\frac{\partial V}{\partial x}) + 1) + ax \frac{\partial V}{\partial x}
\end{aligned}$$

$$(c) -\frac{\partial V}{\partial t} = H^*(u, \frac{\partial V}{\partial x}) = e^{-\rho t} (\rho t + \ln(-\frac{\partial V}{\partial x}) + 1) + ax \frac{\partial V}{\partial x}$$

$$\text{for } V(x, T) = \bar{\Phi}(x) = 0$$

$$(d) V(x, T) = -e^{-\rho T} (c(T) + d(T) \ln(x_T)) = 0 \quad \therefore c(T) = d(T) = 0$$

$$\frac{\partial V}{\partial t} = \rho e^{-\rho t} (c + d \ln(x)) - e^{-\rho t} (\dot{c} + \dot{d} \ln(x))$$

$$\frac{\partial V}{\partial x} = -e^{-\rho t} d \frac{1}{x}$$

$$\begin{aligned}
\therefore H^*(u, \frac{\partial V}{\partial x}) &= e^{-\rho t} (\rho t + \ln(-\frac{\partial V}{\partial x}) + 1) + ax \frac{\partial V}{\partial x} \\
&= e^{-\rho t} (\rho t - \rho t + \ln(d) - \ln(x) + 1) - e^{-\rho t} ad \\
&= e^{-\rho t} (1 - ad + \ln(d) - \ln(x))
\end{aligned}$$

$$\therefore e^{-\rho t} (-\rho c - \rho d \ln(x) + \dot{c} + \dot{d} \ln(x)) = e^{-\rho t} (1 - ad + \ln(d) - \ln(x))$$

$$\therefore \dot{c} = \rho c + \ln(d) - ad + 1$$

$$\dot{d} = \rho d - 1$$

$$\therefore \dot{d} - \rho d = -1$$

$$\text{let } I(t) = e^{\int -\rho dt} = e^{-\rho t}$$

$$\therefore \frac{d}{dt} (e^{-\rho t} d) = -e^{-\rho t}$$

$$\therefore e^{-\rho t} d = \frac{1}{\rho} e^{-\rho t} + c \quad \therefore d = \frac{1}{\rho} + \frac{c}{e^{-\rho t}}$$

$$\therefore d(T) = 0 \quad \therefore \frac{1}{\rho} + \frac{C}{e^{-\rho T}} = 0 \quad \therefore C = -\frac{1}{\rho} e^{-\rho T}$$

$$\therefore d = \frac{1}{\rho} (1 - e^{-\rho T} e^{\rho t}) = \frac{1}{\rho} (1 - e^{-\rho(T-t)})$$

$$\therefore \dot{c} = \rho c + \ln\left(\frac{1}{\rho} (1 - e^{-\rho T} e^{\rho t})\right) - \frac{a}{\rho} (1 - e^{-\rho T} e^{\rho t}) + 1$$

$$\dot{d} = -e^{-\rho T} e^{\rho t}$$

$$(e) \quad u^* = -\frac{e^{-\rho t}}{\partial V / \partial x} = \frac{x}{d}$$

(f) see MATLAB

$$(iii) \quad \text{if } T \rightarrow \infty, \quad d = \frac{1}{\rho}$$

$$\therefore u^* = \rho x$$

$$\therefore H^*\left(u, \frac{\partial V}{\partial x}\right) = e^{-\rho t} \left(1 - ad + \ln(d) - \ln(x)\right)$$

$$= e^{-\rho t} \left(1 - \frac{a}{\rho} + \ln\left(\frac{1}{\rho}\right) - \ln(x)\right)$$